Meet Vamja

cs25mtech14009@iith.ac.in | meetvamja007@gmail.com | +91 9545333242 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.59
St. Lawrence High School	Class XII (Maharashtra State Board)	2020	84.31%
R.P. Vidyalaya	Class X (Maharashtra State Board)	2018	88.60%

PROJECTS

Budget-Aware Multi-LLM Query Routing | *Ongoing Research* | *Guide: Prof. Sandipan Dandapat*

- Designing an adaptive **multi-LLM routing framework** that **decomposes complex user queries** into sub-queries structured as a **directed acyclic graph (DAG)** and assigns each to the **most cost-effective language model** while maximizing overall **response quality**.
- Formulating the routing problem as a **Multiple Choice Knapsack Problem (MCKP)** and implementing a **dynamic programming** based optimizer for **budget-constrained model selection.**
- Building an **end-to-end pipeline** integrating **query decomposition, budget-aware routing**, and **response aggregation** to optimize the **cost-quality trade-off** across heterogeneous LLMs.

RAG Evaluation & Retrieval Engine | *Python, FastAPI, PostgreSQL, ChromaDB, BM25, BGE* **(GITHUB)**

- **Implemented a Retrieval-Augmented Generation (RAG) engine** with asynchronous document ingestion, hybrid retrieval, configurable chunking, and an evaluation-driven pipeline for data-backed architecture decisions.
- Engineered **multi-format document** ingestion (PDFs, tables, OCR scanned pages) with fixed-size, recursive, and semantic chunking via a config-driven strategy registry **measured recursive chunking outperforming semantic chunking by 62.5% on Recall@1.**
- Designed a hybrid retrieval pipeline combining **BGE dense embeddings, BM25 sparse retrieval, Reciprocal Rank Fusion, and cross-encoder re-ranking** achieved **14.8% higher answer** relevance and **8.6% lower hallucination rate** over the un-reranked baseline.

Multi-Agent LLM Program Synthesis | *Python, OpenAI/Gemini APIs, RAG, Prompt Engineering* **(GITHUB)**

- Architected a **4 agent program synthesis framework** with specialized **retrieval, planning, coding,** and **debugging agents** enabling modular generation and iterative self-correction of LLM-generated programs.
- Improved **GPT-4 HumanEval pass@1 from 85.98% to 93.90%** (**7.92-point gain**) through exemplar retrieval, confidence-ranked planning, and test-driven code repair.
- Engineered an automated **validation and self-correction loop** that executes generated code against sample I/O with timeout enforcement and feeds structured failure traces to the debugging agent for iterative repair.
- Built a **modular benchmarking framework** supporting various datasets, multiple LLM providers, and 6 programming languages, with token telemetry, retry tracking, and JSONL-based checkpoint/resume.

Distributed URL Shortener | *Python, C++, FastAPI, Redis, PostgreSQL, Docker* **(GITHUB)**

- Designed a horizontally-scalable URL shortener architecture (sharded PostgreSQL, Redis caching, load balancing) and **verified correctness and reliability through local load testing with zero request failures.**
- Engineered **C++ hot-path extensions** for **Bloom filter collision** checks and FNV-1a consistent-hash routing using pybind11, **reducing P95/P99 redirect latency by 39%/40%** versus the pure-Python implementation under identical load.
- Designed a 150-virtual-node **consistent hashing shard router** to replace modulo-based routing, **reducing key remapping from 74.8% to 38.6%** during **3 to 4 shard expansion**, nearly halving data movement.
- Built **Redis-based read through caching and sliding window rate limiting**, eliminating unnecessary PostgreSQL lookups for non-colliding aliases and achieving **faster collision checks** than direct PostgreSQL queries.

CloudDrive: Distributed File Storage & Sharing Platform | *Python, FastAPI, PostgreSQL, SQLAlchemy* **(GITHUB)**

- Built a **Google Drive style file storage platform** with **resumable uploads** (4 MiB chunks, files up to 2 GB). Interrupted uploads resume by re-sending only the missing chunks.
- Designed a **replicated storage backend** that places each chunk on 2 of 3 nodes using rendezvous hashing, with

SHA-256 integrity checks and automatic replica fallback. reads and writes continue with zero downtime through a full node failure.
- Cut filename search **latency from 3.4 ms to 0.2 ms** on 15,000 files using a **pg_trgm GIN index** and keyset pagination, verified with EXPLAIN ANALYZE.

SKILLS

Programming Languages: C/C++, Python

Technologies: SQL, PostgreSQL, Redis, ChromaDB, Docker, Git

Libraries: FastAPI, Node.js, Pandas, NumPy, Matplotlib, pybind11, sentence-transformers

AI/ML: RAG, Prompt Engineering, BM25, BGE, Cross-Encoder Reranking

ACHIEVEMENTS

Solved **1,500+** Algorithmic problems across LeetCode, Codeforces, and GeeksforGeeks, strengthening algorithmic thinking and optimization skills.

RELEVANT COURSES

Systems: Advanced Computer Networks, Network Security

AI/ML: Foundations of Machine Learning, Deep Learning, Visual Computing

Theory: Advanced Data Structures, Linear Optimization